

STAGE S0

Problem Definition & Need Identification

Consolidated Evidence — Steps 1 through 7, in working order

Team No.	13
Theme No.	Data Processing and Analysis System
Artefact Name:	Distributed Event Processing & Notification Engine
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1. Background & Motivation

Hospitals continuously generate different types of operational events such as patient admissions, doctor availability updates, bed allocation changes, ambulance dispatch requests, blood requirements, appointment scheduling, and emergency alerts. In many hospitals, these events are handled by different departments using separate registers, spreadsheets, or software systems. As a result, staff often spend significant time communicating with multiple departments to obtain the latest information. During emergency situations, delays in receiving critical updates can affect coordination and decision-making. The proposed Distributed Event Processing & Notification Engine aims to address this issue by collecting events from different hospital departments, processing them centrally, and delivering notifications to the appropriate stakeholders in real time. This approach improves information flow, reduces manual communication, and helps hospital staff respond more efficiently to operational and emergency situations.

2. Project Objectives

1. To study the existing process of managing hospital information and identify difficulties in handling information from different services.
2. To develop a centralized system for managing patient details, doctor availability, bed availability, ambulance status, appointments, and blood availability.
3. To process updates from different hospital services and provide the latest information to the concerned staff.
4. To provide notifications for important events such as emergency cases, bed availability, doctor availability, ambulance requests, and blood requirements.
5. To reduce the manual effort involved in checking and updating hospital information and improve the flow of information.
6. To apply Computer Science concepts such as database management, web application development, event processing, notifications, and data analysis in a practical healthcare system.

1. Foundational Courses

Sl.No.	Course	Core Concepts
1.	Database Management System	Patient records, doctor details, bed availability, ambulance details, blood availability, SQL queries, database design.
2.	Software Engineering	Requirements gathering, stakeholder analysis, SDLC, and system design.
3.	Web Technologies	HTML, CSS, Bootstrap, React, and web interface development.
4.	Data Structures	Searching, sorting, filtering, and managing hospital records.
5.	Programming using JavaScript / Node.js	Backend APIs, login, event processing, data processing, and notifications.
6.	Operating Systems	Process management, resource management, file management, and handling different system processes.

2. Existing Problem / Current Situation

Hospitals handle different types of information every day. This includes patient details, doctor availability, bed availability, ambulance status, appointments, and blood availability. This information may be maintained in different registers, spreadsheets, or separate software systems. Hospital staff may need to check different sources or contact different departments to get the required information. During an emergency, staff may need to separately check whether a bed, doctor, ambulance, or required blood is available. Updates from one department may not be immediately available to other concerned staff. There is a need for a system that can collect these updates, process them, and notify the concerned staff. Another challenge is that multiple events can occur simultaneously within the hospital. For example, an emergency patient may require a doctor, bed, ambulance support, and blood availability at the same time. Managing these events manually makes coordination difficult and increases the possibility of delays, missed updates, or communication gaps between departments.

3. Who Is Affected

- ☐ Patients – May face delays in getting information about doctors, beds, appointments, or emergency services.
- ☐ Doctors – Need timely information about emergency cases and patient requirements.
- ☐ Nurses / Hospital Staff – May spend time checking and updating information from different sources.
- ☐ Ambulance Staff – Need timely information about emergency requests and hospital requirements.
- ☐ Blood Bank Staff – Need to update and provide information about blood availability.
- ☐ Hospital Administration – Needs an overall view of hospital resources and ongoing activities.

4. Why the Existing Approach Is Inadequate

- ☐ Information is spread across different sources.
- ☐ Staff may have to check or contact different departments individually.
- ☐ Emergency staff may not get bed, doctor, ambulance, and blood availability in one place.
- ☐ Updates may not reach the concerned staff immediately.
- ☐ Manual communication can take additional time.
- ☐ Staff may have difficulty knowing the latest status of hospital resources.
- ☐ There may be no automatic notification when an important event occurs.
- ☐ It becomes difficult to coordinate different departments during an emergency situation.

5. Evidence That This Problem Is Real

- ☐ Hospitals manage information through multiple departments such as emergency, doctors, beds, ambulance services, and blood banks.
- ☐ Different departments need to share updates about availability and emergency requirements.
- ☐ Emergency situations require quick communication and updated information between departments.
- ☐ Existing hospital management systems show the need to manage patient records, appointments, beds, doctors, and other hospital resources.
- ☐ Our team identified this problem through study of hospital management workflows and existing healthcare system practices.

6. Sources / References

- ☐ Study of existing Hospital Management System workflows.
- ☐ Review of hospital management software and healthcare information systems.
- ☐ Academic study and discussion of hospital emergency and resource management.
- ☐ Project discussions with our project guide.
- ☐ General study of distributed event processing and notification systems.

1. Stakeholder Table

Stakeholder	Role / Description	Needs & Expectations	Current Pain Points	Influence (H/M/L)
Hospital Staff	Handles patient information and emergency requests	Quickly check resources and send notifications	Need to contact different services for information	H
Doctors	Handles patient treatment and responds to emergency requests	Receive notifications and update availability	May not receive updates quickly	H
Ambulance Staff	Handles ambulance requests and availability	Receive requests and update ambulance status	Receive requests and update ambulance status	M
Blood Bank Staff	Manages blood availability	Receive blood request availability	Need to provide availability when requested	M
Patients	Receive hospital services	Timely treatment and service	May face delays during emergencies	M
Admin / Hospital Administration	Manages the hospital system and monitors overall activities	View hospital status, manage users, and monitor resources	Information is spread across different services	H

2. Primary vs. Secondary Stakeholders

Primary Stakeholders	Secondary Stakeholders
Hospital Staff	Admin / Hospital Administration
Doctor	Patients
Ambulance Staff	
Blood Bank Staff	

3. Key Insight Summary

The system should mainly focus on Hospital Staff, Doctors, Ambulance Staff, and Blood Bank Staff. They need quick notifications and easy ways to update availability. The Admin needs to monitor overall hospital activities and resources. Patients can benefit from faster coordination and reduced delays.

1. Current Process — Narrative Summary

When an emergency patient arrives, Hospital Staff collects the patient details and checks the required resources. They contact the Doctor, Bed Staff, Ambulance Staff, and Blood Bank Staff separately to check availability. Each staff member provides the required information. The Hospital Staff then coordinates the available resources for the patient.

2. Step-by-Step Workflow Table

Step No.	Actor / Role	Action Performed	Tool / System Used	Pain Point Observed
1	Hospital Staff	Receives emergency patient and records details	Register / Existing system	Manual entry
2	Hospital Staff	Checks bed availability	Phone / Register / System	Information may be separate
3	Hospital Staff	Checks doctor availability	Phone / Communication	Takes time
4	Hospital Staff	Checks ambulance availability	Phone / Communication	Separate communication
5	Hospital Staff	Checks blood availability if required	Phone / Blood Bank	Manual checking
6	Doctor / Ambulance / Blood Bank Staff	Responds with availability	Phone / Communication	Response may be delayed
7	Hospital Staff	Collects responses and coordinates required resources	Manual process	No single view of updated information

3. Workflow Diagram

Emergency Patient Arrives
 ↓
Hospital Staff Records Patient Details
 ↓
Check Doctor Availability
 ↓
Check Bed Availability
 ↓
If Required → Check Ambulance / Blood Availability
 ↓
Receive Responses from Staff
 ↓
Coordinate and Provide Treatment

4. Bottlenecks & Inefficiencies Identified

- ☐ Information is available in different places.
- ☐ Staff need to contact different departments separately.
- ☐ Availability responses may be delayed.
- ☐ Manual communication takes extra time.
- ☐ There is no single place to see all current statuses.
- ☐ Important updates may not reach the concerned staff immediately.

1. Observed Problems → Underlying Needs

Observed Problem	Evidence / Source	Underlying Need	Linked Stakeholder(s)
Hospital information is in different places	Study of current hospital workflow	Central access to information	Hospital Staff, Admin
Staff need to contact different departments for availability	Current workflow study	Quick communication between staff	Hospital Staff, Doctor, Ambulance Staff, Blood Bank Staff
Emergency updates may take time to reach staff	Current workflow study	Timely updates	Hospital Staff, Doctor
Bed availability is not available in one place	Current workflow study	Quick bed status updates	Hospital Staff, Admin

Observed Problem	Evidence / Source	Underlying Need	Linked Stakeholder(s)
Ambulance and blood availability need separate checking	Current workflow study	Quick resource availability updates	Hospital Staff, Ambulance Staff, Blood Bank Staff
There is no single view of current resource status	Review of hospital workflow	Updated overall status	Hospital Staff, Admin

2. Prioritization

Priority	Need	Reason
1	Quick emergency communication	Important during emergencies
2	Timely availability updates	Helps staff respond quickly
3	Central access to information	Reduces separate checking
4	Updated bed status	Important for patient admission
5	Ambulance and blood status	Needed when required
6	Overall resource view	Helps administration monitor activities

3. Problem Statement

Hospitals generate a large number of operational events every day, including patient admissions, doctor availability updates, bed allocation changes, ambulance requests, blood availability updates, appointment scheduling, and emergency alerts. These events are often managed by different departments using separate systems or manual communication methods, making it difficult for hospital staff to receive accurate and timely information. During emergency situations, delays in sharing critical updates can affect coordination between departments and slow down patient care. Therefore, there is a need for a Distributed Event Processing & Notification Engine that can collect events from multiple hospital departments, process them in real time, and deliver notifications to the appropriate stakeholders through a centralized platform.

4. Project Objectives

- 1. To develop a centralized event processing platform for managing hospital operational events from different departments.**
- 2. To provide real-time updates on doctor availability, bed occupancy, ambulance status, blood availability, and emergency requests.**
- 3. To automatically process and distribute hospital events to the appropriate staff based on their roles and responsibilities.**
- 4. To reduce manual communication and improve coordination between hospital departments through timely notifications.**
- 5. To maintain an event history and notification log for tracking important hospital activities and resource updates.**
- 6. To provide hospital administrators with a centralized dashboard for monitoring hospital resources, ongoing events, and emergency operations.**

1. In-Scope

The proposed system focuses on developing a centralized Distributed Event Processing & Notification Engine for hospital operations. It includes patient registration, doctor availability management, bed status updates, ambulance request management, blood availability monitoring, emergency event processing, and real-time notification delivery. The system processes events generated by different hospital departments, stores them in a centralized database, and sends notifications to authorized users based on their roles. It also provides an administrative dashboard to monitor hospital resources, event status, and notification history using sample hospital data.

2. Out-of-Scope

The project is limited to hospital event processing, resource monitoring, and notification management. It does not include online payment or billing systems, pharmacy inventory management, medicine ordering, AI-based medical diagnosis, laboratory report generation, video consultation, wearable or IoT medical device integration, or integration with real hospital information systems. SMS, WhatsApp, or external government healthcare integrations are also outside the scope of this project. These features require external infrastructure, advanced medical systems, or additional development beyond the semester project timeline.

3. Assumptions

- ☐ **Hospital staff, doctors, ambulance staff, blood bank staff, and administrators will have authorized access to the system through role-based login accounts.**
- ☐ **Each department will update its resource availability and operational events whenever a change occurs.**
- ☐ **Basic patient, doctor, bed, ambulance, and blood availability information will be maintained using sample data.**
- ☐ **The system will process events generated within the application and deliver notifications to the relevant stakeholders.**
- ☐ **Internet or local network connectivity will be available for communication between different modules of the system.**

4. Constraints

- ❑ The project is developed by a student team within the available semester duration and limited development resources.
- ❑ Real hospital databases and live hospital information systems are not available for testing.
- ❑ The application will use sample hospital data instead of real patient records.
- ❑ Real-time notifications will be demonstrated within the web application and not through external messaging services.
- ❑ The project scope is limited to core event processing, notification management, and resource monitoring features to ensure successful implementation within the project timeline.

1. Technical Feasibility

Aspect	Assessment	Justification
Technology	Feasible	We can use HTML, CSS, JavaScript, Node.js, and MySQL.
Development Skills	Feasible	The team has basic programming and database knowledge.
Notification System	Feasible	Notifications can be implemented using web-based technologies.
Hardware & Software	Feasible	Development can be done using available computers and free software tools.

2. Operational Feasibility

The system is feasible because hospital staff can use simple role-based screens to view requests, receive notifications, and update availability.

3. Time Feasibility

The project is achievable within 15 weeks because the scope is limited to patient details, resource availability, emergency requests, notifications, and basic administration.

4. Resource Feasibility

The project can be developed using the team's available computers, programming tools, and sample hospital data. No special hardware or real hospital data is required.

5. Overall Feasibility Conclusion

Go with modifications. The project is feasible within the semester if we keep the scope limited and use sample data instead of integrating with real hospital systems.

1. Project-Level Risks

Risk ID	Risk Description	Category	Likelihood (H/M/L)	Impact (H/M/L)	Mitigation Strategy
R1	Team may have difficulty implementing notifications	Technical	M	H	Start notification development early and test with sample events.
R2	Team may have limited knowledge of some technologies	Technical	M	M	Learn required concepts and divide tasks based on skills.
R3	Project scope may become too large	Scope	M	H	Keep only the planned features in the project.
R4	Development may take more time than expected	Timeline	M	H	Divide the work into weekly tasks and track progress.
R5	Sample hospital data may be limited	Resource	M	M	Use suitable sample data for development and testing.
R6	Database errors may cause incorrect information	Technical	M	H	Validate data and test database operations regularly.

Risk ID	Risk Description	Category	Likelihood (H/M/L)	Impact (H/M/L)	Mitigation Strategy
R7	Team members may not complete assigned tasks on time	Resource	M	M	Assign clear responsibilities and review progress regularly.
R8	Integration between different modules may cause problems	Technical	M	M	Develop and test modules step-by-step before final integration.

2. Risk Owner Summary

Each team member will monitor the risks related to their assigned work. The project coordinator will track scope, timeline, and task completion, while technical members will monitor development, database, notification, and integration risks. The team will review these risks regularly and take corrective action when required.